Counterfactual Preference Perturbation Localises Objective Misalignment in Multi-Agent LLM Negotiation

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Abstract

In a multi-agent LLM system, an agent can satisfy its declared objective at the outcome level without that objective having been load-bearing in producing its behaviour. We introduce *counterfactual preference perturbation*, a post-hoc auditing methodology that takes behavioural logs and declared reference objectives as its only inputs and asks whether each agent's behaviour was coupled to its declared preferences. The framework holds the logs fixed and interpolates one agent's score sheet from its declared preferences toward a misaligned target, measuring the response through Partial Information Decomposition. If the behaviour was coupled to the reference, synergy in pairs involving that agent drops as the perturbation grows. If not, the response is flat. Applied to a controlled mismatch experiment on a multi-agent LLM negotiation benchmark, the perturbation response identifies which agents played with modified preferences. The framework localises objective mismatch across three orthogonal axes: by agent, by preference dimension, and by time window. A cross-environment probe on a common-pool resource scenario recovers the misalignment signal in the mismatch condition, though with a residual response in the control that reflects substrate-dependent encoding. We present this work as proof of principle for information-theoretic auditing of multi-agent LLM behaviour from logs and declared objectives alone.

Keywords

Multi-Agent LLM Systems, Partial Information Decomposition, Multi-Agent Safety, Counterfactual Auditing, AI Alignment

1 Introduction

In a multi-agent LLM system, an agent can satisfy its declared objective at the outcome level without that objective having been load-bearing in producing its behaviour. An auditor reviewing such an interaction must therefore ask, for each agent, whether the observed actions were actually generated by the declared objective or merely produced compatible outcomes for other reasons. This question is not *attributable* from terminal-state metrics. Deal rates, welfare scores, and payoff distributions can register that an interaction went wrong, but not how the alignment structure was organised or which agent's objective was load-bearing in producing it. The limits of outcome-only metrics for capturing multi-agent dynamics have been observed empirically across collaborative-reasoning and deliberative settings [6, 12].

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Throughout, we use *objective misalignment* to mean a mismatch between an agent's declared reference objective and the objective that actually generated its observed behaviour, without any claim about the agent's intent or whether the mismatch was strategically pursued.

Existing approaches to multi-agent auditing fall along three lines. White-box methods train probes on agent activations to detect coordinated misalignment [15], which requires inference-time access to model internals. Communication-channel methods intervene on the messages agents exchange, through paraphrase, rate limiting, or role permutation [16]. Outcome-based audits measure deviation from cooperative optima through regret or welfare metrics [9]. Each captures a real failure mode but leaves a methodological gap: none provides a behaviour-level account of how observed joint behaviour depends on agents' declared objectives, operating only on logs and reference preferences without model access or channel intervention.

We propose *counterfactual preference perturbation*, a post-hoc auditing methodology that fills this gap. Given behavioural logs and a known reference objective for each agent, the framework holds the logs fixed and interpolates one agent's score sheet from its declared preferences toward a misaligned target. The response is measured via Partial Information Decomposition, which decomposes the mutual information between two agents' behavioural states and a shared outcome into redundant, unique, and synergistic components. Synergy captures information that only the joint observation of both agents' states reveals. If an agent's behaviour was coupled to its declared preferences, perturbing those preferences should disrupt pair synergy involving that agent. If the behaviour was decoupled from those preferences, the perturbation should produce no response. The key empirical finding is that the *level* of PID synergy carries no signal about objective mismatch on its own; only the *response* to counterfactual perturbation does.

The framework localises misalignment along three orthogonal axes: which agent's behaviour is inconsistent with its declared objective, which preference dimension carries the inconsistency, and over which time window the inconsistency holds. We validate each localisation claim on the multi-agent negotiation benchmark of Abdelnabi et al. [1] with cluster-bootstrap confidence intervals at the game level. The dimension axis is demonstrated when the auditor targets a candidate dimension; a fully blind scan over all agent-by-issue cells surfaces the ground-truth dimension among the largest responses but does not on its own isolate it (Section 4.1). We further probe transferability on two additional substrates. On GovSim, a common-pool resource scenario [11], the framework recovers the outsider in a mismatch condition but exhibits residual response in the control, consistent with substrate-dependence of the

source-feature encoding (Section 4.3, Appendix H). On an English auction benchmark, the perturbation response does not localise despite a behaviourally manifest manipulation, which we report as a documented scope condition (Appendix F).

We present this work as proof of principle for an information-theoretic approach to multi-agent behaviour auditing. Two contributions support this framing. First, we introduce counterfactual preference perturbation as a post-hoc methodology that operates on behavioural logs and declared reference objectives. Second, we demonstrate localisation along three orthogonal axes within a single framework: by agent, by time window, and by preference dimension. The agent and time-window axes are cleanly validated with cluster-bootstrap confidence intervals on a negotiation benchmark; the dimension axis is demonstrated when the auditor targets a candidate dimension, while a fully blind scan recovers it only partially. The broader claim is that information-theoretic decomposition of agent behaviour under counterfactual perturbation captures structural properties of multi-agent interaction that outcome-level measurement does not, and that this approach merits further development across substrates, intervention objects, and detection targets.

2 Related Work

This work sits at the intersection of multi-agent LLM safety auditing and information-theoretic analysis of agent behaviour. We discuss four lines of prior work that the paper builds on or contrasts with.

Multi-agent LLM safety and collusion auditing. The risks posed by multi-agent LLM systems have been articulated as a structured research agenda by Hammond et al. [3], who call for new methods of evaluating cooperative capabilities, vulnerabilities, and behaviours that emerge from agent interactions rather than relying on existing single-agent evaluation paradigms. This work contributes to that agenda by introducing a behaviour-level method for localising objective misalignment in multi-agent LLM systems. The inadequacy of outcome-only metrics for capturing the dynamics of multi-agent LLM systems has been observed empirically across collaborative-reasoning and deliberative settings [6, 12], and motivates the structural approach we develop here. Motwani et al. [7] formalise the problem of secret collusion among generative AI agents and provide a model-evaluation framework for steganographic coordination. Tailor [16] proposes capacity-guided auditing through interventions on the agents' communication channel, such as paraphrase, rate-limiting, and role-permutation. Rose et al. [15] detect collusion by learning a deception direction from agent activations and aggregating per-agent scores into group-level statistics under a distributed anomaly detection framing. The approach requires white-box activation access to all agents during inference, an assumption the authors note holds for internal deployments and pre-deployment evaluations but not for third-party auditing of closed-weight systems. The framework presented here takes behavioural logs and declared preference sheets as its only inputs. This positions the method for post-hoc auditing of completed multi-agent interactions, including forensic review of incidents and compliance assessments where re-running inference with model access is not available.

Outcome-based evaluation of multi-agent LLM systems. Several benchmarks measure cooperation and welfare in multi-agent LLM settings. Welfare Diplomacy [8] reformulates Diplomacy as a general-sum game to measure cooperative competence. GovSim [11] measures sustainability under commons dilemmas, finding that only a small fraction of tested models achieves sustainable outcomes. Colosseum [9] grounds cooperation as a distributed constraint optimisation problem and audits collusion through regret relative to the cooperative optimum and the coalition-advantage gap, arguing that judging message logs alone is insufficient because agents may discuss collusion without acting on it. The task we address is distinct: localising, within a single interaction, whether each agent's behaviour is consistent with its declared objective. In our controlled experiment that question is not recoverable from outcome-level measurement alone, motivating the behaviour-level approach we develop.

Counterfactual interventions for LLM auditing. Khanzadeh [5] proposes Project Ariadne, a structural-causal framework that intervenes on intermediate reasoning nodes of single-agent LLMs to measure causal sensitivity and identify a "causal decoupling" failure mode in which terminal outputs are insensitive to upstream reasoning. The framework presented here extends this counterfactual-intervention logic from single-agent reasoning traces to multi-agent reference preferences, and replaces causal sensitivity with an information-theoretic response that decomposes coordinated coupling between pairs of agents. The auditor's hypothesis about an agent's objective is the perturbed variable, behavioural logs are held fixed, and the response is measured through changes in pair synergy.

Information-theoretic analysis of agent interactions. Partial information decomposition [17] separates the mutual information that two sources provide about a target into redundant, unique, and synergistic atoms. Williams and Beer's $I_{\min}$ estimator is known to overestimate redundancy in high-dimensional source spaces, which Bertschinger et al. [2] address through the BROJA constrained-optimisation alternative. We use $I_{\min}$ as the primary estimator and verify that the headline response is preserved under BROJA on aggregated condition distributions. The closest application of PID to multi-agent LLMs is Riedl [13], who uses partial information decomposition of time-delayed mutual information to detect emergent synergy in a guessing-game setting without inter-agent communication.

3 Methods

3.1 Partial Information Decomposition

Mutual information $I(X;Y)$ quantifies, in bits, the average reduction in uncertainty about Y obtained by observing X. When two source variables S_1, S_2 jointly carry information about a target T, the joint quantity $I(S_1, S_2; T)$ pools contributions of qualitatively different kinds: some information sits in either source alone, some is shared between them, and some is accessible only when the pair is observed together. Partial Information Decomposition (PID) [17] makes this distinction explicit by writing

$$I(S_1, S_2; T) = \mathrm{Red} + \mathrm{Un}_1 + \mathrm{Un}_2 + \mathrm{Syn}, \tag{1}$$

where each *atom* on the right-hand side is non-negative. *Redundancy* (Red) is information about T that either source provides on its own; *unique* information (Un_i) is what S_i contributes that S_{3-i} does not; and *synergy* (Syn) is the part of the joint mutual information present only in the pair — neither source alone carries it, and it cannot be recovered by inspecting the sources separately. The canonical example is $T = S_1 \oplus S_2$ with independent binary sources: each S_i alone is uninformative about T, yet the pair determines it, so the entire joint mutual information is synergistic.

For our framework, synergy is the load-bearing atom. Two agents whose behavioural states are conditionally independent given the system outcome contribute no synergy, even if each state is individually predictive of that outcome; synergy is non-zero only when the joint state encodes information that neither state encodes alone. Synergy therefore tracks *coupling* between the sources rather than their individual informativeness — which is precisely what a counterfactual perturbation needs to act on. If behaviour is genuinely organised around an agent's reference preferences, perturbing those preferences should disrupt the joint structure that synergy measures; redundant and unique information, by construction, need not respond in the same way.

We estimate the decomposition using the Williams-Beer $I_{\min}$ definition [17], which assigns redundancy as

$$I_{\min}(S_1, S_2; T) = \sum_{t \in \mathcal{T}} p(t) \min_{i \in \{1,2\}} I(T = t; S_i), \tag{2}$$

where

$$I(T = t; S_i) = \sum_{s_i} p(s_i \mid t) \log_2 \frac{p(t|s_i)}{p(t)}$$

is the specific information S_i provides about $T = t$. The $I_{\min}$ estimator is known to over-attribute redundancy in some regimes [2]. Appendix D confirms that $I_{\min}$ and BROJA agree within 4% across conditions in our experiments. We prefer $I_{\min}$ over BROJA for tractability. BROJA solves a convex optimisation per joint distribution and does not scale to the cell counts and cluster-bootstrap budgets we use.

We apply PID pairwise to all $\binom{6}{2} = 15$ agent pairs. Pairwise resolution is necessary rather than convenient: a system-level information-theoretic measure such as O-information [14], which aggregates over the full six-agent state, shows no consistent response to perturbation across games (Section 4.2, Figure 7), whereas the pairwise decomposition both responds reliably and identifies *which* partner relationships carry the alignment-relevant signal, not only *which* agent is affected. We initially used the `dit` library [4], then re-implemented $I_{\min}$ directly in NumPy to reduce per-iteration overhead for the bootstrap analyses.

3.2 Counterfactual Preference Perturbation

The framework targets a specific question: was an agent's observed behaviour coupled to its declared preferences? A direct test holds the behavioural logs fixed and asks how the information structure responds when those preferences are systematically altered. If the agent acted on the reference sheet, perturbing it should disrupt the synergy in pairs containing that agent; if the agent's behaviour was decoupled from the sheet, no response should appear.

Feature encoding. Each agent i's source at round t combines two scalar features: *self-score*

$$\sigma_i^t = \mathtt{score}_i(d^t),$$

the payoff agent i receives under the shared-table proposal d^t at round t; and *payoff sensitivity*

$$\psi_i^t = |\mathtt{score}_i(d^t) - \mathtt{score}_i(d^{t-1})|/100,$$

the magnitude of change in i's payoff between consecutive shared-table proposals. Self-score captures what the agent is currently getting. Sensitivity captures how responsive its payoff is to the ongoing proposal stream. Each feature is quantile-binned into $B = 4$ bins, yielding $B^2 = 16$ joint states per agent.

Bin edges are fitted per condition at $\alpha = 0$, not pooled across conditions. The experimental design produces a distributional shift between conditions by construction (aligned vs. misaligned behaviour distributes self-score and sensitivity differently). Aggregate quantiles do not represent within-condition distributions, and pooling produces biased binning that suppresses the perturbation signal. We discovered this empirically in a temporal-localisation extension (Section 4.1) and apply per-condition edges throughout.

The target for pair (j, k) is the non-pair welfare delta

$$\Delta^t_{\setminus jk} = \sum_{a \notin \{j,k\}} \left[\mathtt{score}_a(d^{t+1}) - \mathtt{score}_a(d^t)\right], \tag{3}$$

binned into three terciles. This choice isolates how the pair's joint state predicts the externality on the remaining four agents, rather than how it predicts the pair's own payoffs. Synergy on a self-referential target would conflate genuine coupling with mechanical accounting. The resulting joint table has $16 \times 16 \times 3 = 768$ cells, estimated from approximately $25 \times 21 \approx 525$ (σ, ψ, Δ) triplets per pair per condition across the game population.

Counterfactual perturbation. For a designated *drift agent* i, we hold all observed behaviour fixed and interpolate i's score sheet from the reference toward a fully inverted copy:

$$\mathtt{scores}_i(\alpha) \;=\; (1-\alpha)\,\mathtt{scores}_i^{\mathrm{ref}} + \alpha\,\mathtt{scores}_i^{\mathrm{inv}},\ \alpha \in [0,1]. \tag{4}$$

The inverted sheet $\mathtt{scores}_i^{\mathrm{inv}}$ reverses the within-issue ranking of options: the agent's highest-valued option on each issue becomes its lowest and vice versa, with intermediate options mapped accordingly. Sweeping α from 0 to 1 in 11 steps traces a continuous path from reference to adversarial preferences.

Only agent i's source features are re-encoded under $\mathtt{scores}_i(\alpha)$; all other agents' sources and the welfare target use reference scores throughout. The proposals d^t, the agents' messages, and the realised deal trajectory are unchanged, only the score function used to evaluate them shifts. No new LLM calls are made during the sweep: the same behavioural logs are re-evaluated under different reference encodings. The framework therefore probes encoding-behaviour consistency alone, free of confounds from regeneration or sampling variation.

Drift pairs and controls. For each drift agent i, the 15 agent pairs partition into the five drift pairs containing i and the ten control pairs that do not. We compute PID atoms separately for each pair at each α. If i's behaviour was coupled to its reference preferences, synergy in drift pairs should decline with α. Control pairs, unaffected by the perturbation by construction, should remain flat

and serve as a within-game baseline. We define the perturbation response of pair (j, k) as the synergy ratio

$$r_{(j,k)}(\alpha) = \mathrm{Syn}_{(j,k)}(\alpha)/\mathrm{Syn}_{(j,k)}(0),$$

averaged across drift pairs to summarise agent-level coupling. The headline statistic is the ratio at the perturbation endpoint, averaged over the drift pairs containing agent i,

$$r_i = \frac{1}{|\mathcal{P}_i|} \sum_{(j,k)\in\mathcal{P}_i} r_{(j,k)}(\alpha = 1), \qquad \mathcal{P}_i = \{(j,k) : i \in \{j,k\},\ j < k\}.$$

A value $r_i \ll 1$ indicates behaviour coupled to the reference; $r_i \approx 1$ indicates decoupling.

Confidence intervals. We quantify uncertainty by cluster bootstrap, treating each game as an independent unit and resampling games with replacement (1,000 resamples, 2.5/97.5 percentiles). Within a game, the approximately 21 per-pair triplets share a single negotiation trajectory and are not independent: a coalition forming early or a particular agent dominating influences every subsequent round. A naive bootstrap over triplets ignores this within-game correlation and produces narrow, anti-conservative intervals. Game-level resampling preserves trajectory structure, a resampled game contributes all its triplets together, and yields valid intervals on all reported quantities.

Benchmark and data. We use the LLM-Deliberation benchmark of Abdelnabi et al. [1], in which six LLM agents negotiate over five issues across 24 rounds to reach a joint agreement. Each agent holds a private preference sheet assigning integer scores to options on each issue, together with a minimum acceptable total score below which the agent rejects any deal. Payoffs are deterministic sums of per-issue option scores over a deal space of $3 \times 3 \times 4 \times 4 \times 5 = 720$ possible agreements. Agents communicate in natural language and submit proposals through a structured channel, producing a behavioural log that records each agent's proposals and messages at every round alongside the deal under discussion. The score-sheet structure provides controlled ground truth on agent objectives, making the benchmark well-suited as a calibration substrate for behaviour-auditing methods that require a known reference objective. We use the six released game configurations together with two new datasets generated using GPT-4.1-mini at temperature 0.9: 65 games for the controlled mismatch experiment (Section 4) and 25 games for the temporal-intervention extension in which one agent's preferences are silently swapped mid-game (Section 4.1).

4 Experiments and Results

We organise the results around three questions. Section 4.1 shows that the perturbation response localises the mismatch along three orthogonal axes, by agent, by time window, and by preference dimension. Section 4.2 asks why synergy specifically is the load-bearing measure, comparing it against three scalar alternatives. Section 4.3 probes generality and scope: transfer to a common-pool resource substrate and a documented negative case on a competitive auction.

4.1 Localising Misalignment

As a preliminary check on the six released game configurations, the perturbation produces a selective, stakes-graded response. Across

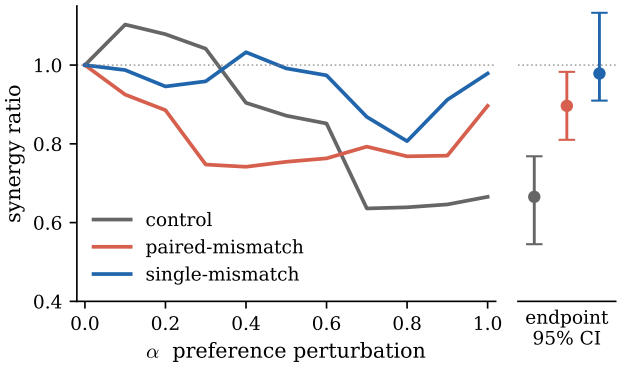


Figure 1: Agent-level localisation. Synergy ratio $r(\alpha) = \mathrm{Syn}(\alpha)/\mathrm{Syn}(0)$ for the probed agent's drift pairs across the perturbation sweep, in three conditions (25 control, 25 paired-mismatch, 15 single). Right panel: cluster-bootstrap 95% CIs on the endpoint ratio at $\alpha = 1$, shared y-axis. Control drops to 0.665 $[0.545, 0.768]$, paired mismatch to 0.896 $[0.810, 0.983]$, single mismatch sits at 0.979 $[0.910, 1.133]$. The control interval separates cleanly from both mismatch conditions; the paired- and single-mismatch intervals overlap.

19 agent-specific profiles (three to four drift agents per game), drift-pair synergy falls 18–72% as α sweeps from reference to inverted, while same-game control pairs remain flat, and the steepness of the drop tracks how much each agent has at stake in the negotiation (Appendix A, Figure 5). This establishes that the response exists, is agent-specific, and is interpretable. What it cannot establish is whether the response tracks *ground-truth* misalignment, since every agent in those games plays honestly under its true preferences. The three experiments below supply that ground truth and test localisation along three orthogonal axes: which agent is misaligned, when the misalignment begins, and which preference dimension carries it.

By Agent. We construct three conditions (25 control, 25 paired-mismatch, and 15 single-mismatch games) in which one or two agents play under a modified score sheet. We designate one of the six agents as the *probed agent* and one of the five negotiation issues as the *target issue*. In the *control* condition all agents play under their original preferences. In the *single* condition the probed agent plays with the target issue reversed, an option ordering swap that flips highest-valued and lowest-valued options. In the *paired-mismatch* condition the probed agent and a second agent both play under reversed target-issue preferences.

The question is whether the perturbation response, measured by the synergy ratio r at the perturbation endpoint, separates these three conditions. The prediction is direct. In control, the probed agent's behaviour is generated by the original preferences, so perturbing those preferences should disrupt the encoding and lower synergy ($r < 1$). In single mismatch, the probed agent's behaviour is generated by the modified preferences, so perturbing the original preferences toward the modified ones should produce no drop ($r \approx 1$).

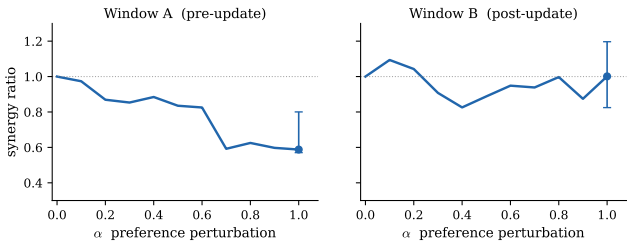


Figure 2: Temporal localisation. Synergy ratio sweep for the probed agent's drift pairs in two windows of a 25-game intervention experiment. The probed agent plays under reference preferences in rounds 1–12 (Window A) and under reversed target-issue preferences in rounds 13–24 (Window B). Per-window bin edges; shaded bands are cluster-bootstrap 95% CIs. Window A ratio 0.588 [0.570, 0.800]**, Window B ratio** 1.001 [0.825, 1.197]**. The intervals do not overlap.**

Figure 1 reports the per-condition ratios. The control ratio is 0.665 with confidence interval [0.545, 0.768], excluding 1.0 in the predicted direction. The single mismatch ratio is 0.979 with interval [0.910, 1.133], overlapping 1.0 and consistent with full decoupling from the original reference. The paired-mismatch ratio is 0.896 with interval [0.810, 0.983], excluding 1.0 below and separating from the control interval.

By Time Window: Mid-Game Intervention. The previous experiment tells us which agent is misaligned given the whole game. A natural follow-up asks whether the framework can localise *when* misalignment begins. We design a temporal intervention. The probed agent plays the first 12 rounds under its original target-issue preferences. At round 12 the score sheet is silently swapped, and the probed agent's reasoning for rounds 13 through 24 now reflects the inverted preferences. The agent itself notices the discontinuity, sometimes flags it verbally, and then proceeds. The experiment produces 25 such games.

We then split each game's log into two windows. Window A covers rounds 1 through 12 (the pre-intervention period). Window B covers rounds 13 through 24. We run the perturbation analysis independently within each window, using per-window bin edges rather than a single global discretisation. This last point matters. The two windows produce different distributions of self-score and sensitivity values by construction, since the agent's reference changed between them. Pooling the windows when fitting bin edges smooths over this distributional shift and suppresses the perturbation signal in whichever window has narrower variance. Per-window edges respect the design.

The result confirms temporal localisation (Figure 2). Window A produces ratio 0.588 with interval [0.570, 0.800], excluding 1.0 in the aligned direction. Window B produces ratio 1.001 with interval [0.825, 1.197], overlapping 1.0. The two intervals do not overlap (0.800 below 0.825), so the windows are statistically distinguishable. The framework finds the change-point: pre-intervention behaviour is coupled to the original reference, post-intervention behaviour is not.

By Dimension. A second follow-up asks which preference dimension the misalignment lives on. In the agent-level experiment we knew to perturb the target issue; in a real audit the dimension is not known in advance.

We first run a per-issue ablation, reversing the probed agent's preferences on one issue at a time using the existing game data. Because the agent-level modification was confined to the target issue, the target-issue ablation reproduces the agent-level result of Section 4.1 (control 0.665, mismatch 0.896); the value of the ablation is the contrast *across* issues. Only the target issue carries the gap: two low-weight issues return null signals in both conditions (insufficient variation for the estimator), and two show only weak, opposite-direction differences (mismatch 0.78 and 0.80 against controls 0.95 and 1.07).

We then run a joint scan across all 30 agent-by-issue cells, the blind setting in which neither the misaligned agent nor dimension is known. The two ground-truth-misaligned cells appear among the largest positive gaps (Figure 3), but so do a few others, so the raw scan does not by itself isolate the misaligned cells; cleanly separating direct misalignment from indirect effects in the blind setting is left to future work.

4.2 Why Synergy: Comparison with Scalar Measures

To justify the use of PID, we compare four candidate metrics across the six released game configurations: the best-response gap (BR-gap), the distance between an agent's observed action and its payoff-maximising response given the others' actions, the Nash bargaining solution distance [10], multivariate O-information [14], and PID synergy. Each is computed at the same alpha values and normalised by its $\alpha = 0$ value. The BR-gap traces a V-shape, dropping and then rising as alpha increases, consistent with a magnitude-based measure that ignores directional information. The Nash bargaining solution does not respond. O-information responds but with high variance and no consistent direction. PID synergy is the only measure with a clean monotonic response (Appendix I, Figure 7). We additionally compute the response in each of the four PID atoms separately and find that across the control and paired-mismatch conditions, 81 to 96 percent of the perturbation response is concentrated in synergy, with redundancy and the two unique atoms remaining stable. The perturbation is not just shifting overall predictability. It is specifically affecting the synergistic component, which is what the framework is designed to measure.

4.3 Generality and Scope

To probe whether the framework transfers beyond multi-issue negotiation, we apply counterfactual preference perturbation to GovSim's fishing commons scenario [11]. Five LLM agents share a regenerating fish stock, and we run 30 games per condition with Llama-3-70B (vLLM, unquantised). In the mismatch condition, one agent, the *outsider*, enters at round 3 with declared preferences that favour maximal extraction over sustainability, while the other four residents hold sustainable preferences. The control condition retains the same population with all five agents sustainable.

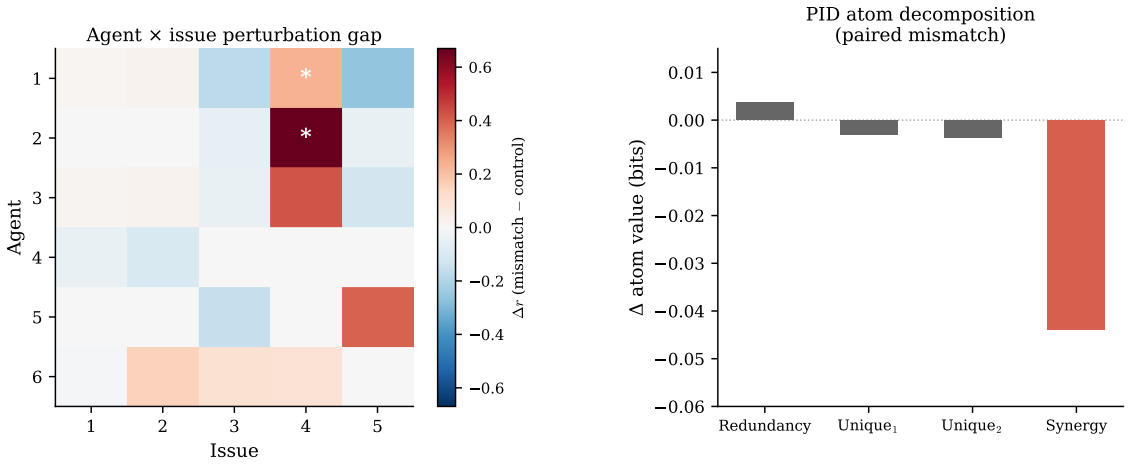


Figure 3: Joint agent-by-issue scan and atom decomposition. Left: gap $\Delta r = r_{\text{mismatch}}(\alpha = 1) - r_{\text{control}}(\alpha = 1)$ across all 30 (agent, issue) cells. Markers (∗) indicate the two ground-truth-misaligned cells (the probed agent and the second agent, both on the target issue), which are among the largest-gap cells in the blind scan. Right: absolute change in atom value (bits) from $\alpha = 0$ to $\alpha = 1$ for the four PID atoms under paired mismatch, drift pairs only. The synergistic atom carries the response; redundancy and the two unique atoms are statistically flat.

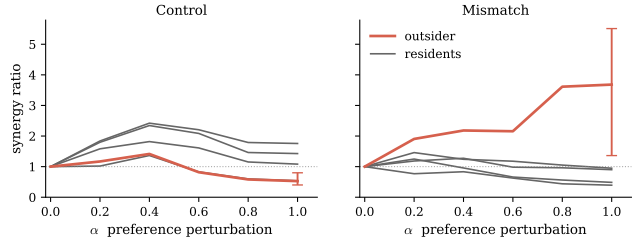


Figure 4: Cross-environment probe: GovSim common-pool resource. Synergy ratio sweep for each of five agents in the control (left) and mismatch (right) conditions of 30 games each, Llama-3-70B. The outsider enters at round 3 in mismatch as an extractive agent; the four residents are sustainable. The outsider's ratio rises monotonically under mismatch to 3.68 [1.36, 5.51] (cluster-bootstrap, errorbar shown), uniquely localising the misaligned agent. The wide interval reflects the outsider's smaller triplet pool (212 vs ∼482 for the residents) as the commons collapses early. Two residents also rise above 1.0 in control, exhibiting residual α-sensitivity that the negotiation encoding does not show; substrate-dependence of feature encoding is discussed in Section 5.

GovSim's per-round harvest decision is one-dimensional and lacks a payoff-sensitivity analogue. The source encoding is therefore substrate-specific. For agent i at round t with harvest h_i^t and sustainable share s_i^t, we define

$$f_{1,i}^t(\alpha) = \frac{h_i^t}{(1-\alpha)\, s_i^t + \alpha \cdot 30}, \qquad f_{2,i}^t(\alpha) = \left| f_{1,i}^t(\alpha) - f_{1,i}^{t-1}(\alpha) \right|. \tag{5}$$

The interpolated target sweeps between the agent's sustainable share and an extractive target of three times sustainable share. The pair target is the non-pair harvest delta, parallel to the negotiation framework and α-independent. Bin edges are fitted per (condition, drift agent) across all α. Full encoding details are documented in Appendix H.

At the outcome level, mismatch reduces mean commons survival from 11.7 rounds to 5.8 and raises outsider harvest from 5.6 to 29.5 fish per round. The misalignment is therefore detectable by outcome metrics. Outcome detection is not the contribution claimed here. We report it to establish that the misalignment is behaviourally manifest in this substrate before turning to localisation.

The perturbation response identifies the outsider directly (Figure 4). Under mismatch, the outsider produces ratio 3.68 with cluster-bootstrap interval [1.36, 5.51], the only agent whose interval excludes 1.0 in the upward direction. Two residents drop as predicted (to 0.39 [0.27, 0.60] and 0.49 [0.36, 0.82]), and two are statistically flat (0.95 [0.74, 1.22] and 0.90 [0.74, 1.26]). This is the framework's expected signature. The outsider's pair synergy rises as α shifts the reference toward extraction because the outsider's behaviour was generated by exactly this kind of preference, while the residents respond to the outsider's behaviour without their own joint structure tracking α. The outsider's triplet pool under mismatch is smaller than the others (212 versus approximately 482 for the residents) because the commons collapses early, which the wide bootstrap interval reflects.

Control behaviour is mixed. Three of the five agents behave as expected (one resident and the outsider drop, one resident is flat), but the remaining two residents produce ratios 1.76 [1.24, 2.43] and 1.43 [1.11, 1.89], intervals that exclude 1.0 in the wrong direction. The same source encoding that cleanly separates conditions in negotiation exhibits residual α-sensitivity in GovSim's distributional structure even when no agent is misaligned. We read this

as substrate-dependence of the feature encoding, not as method failure. Two observations support the reading. First, the directional specificity of the mismatch result is preserved, since only the outsider's interval rises above 1.0. Second, the two misfiring residents no longer rise under mismatch (from 1.76 to 0.90 and from 1.43 to 0.49), which is what one would expect if the substrate noise is dominated by a genuine misalignment signal when one is present.

On a competitive English auction the perturbation response does not localise despite a behaviourally manifest manipulation, a scope condition we document in Appendix F.

5 Discussion

We tested whether counterfactual perturbation of declared preferences elicits a measurable response in the information structure of multi-agent behaviour, and whether that response can localise the agent, the preference dimension, and the time window where an objective mismatch lives. Across three localisation axes and two substrates, pair synergy involving a tested agent moves with the perturbation when the agent's behaviour was generated under preferences resembling the reference, and remains approximately flat when it was not.

Mechanism. The perturbation does not change what the agents did. It changes the encoding through which their actions are read as features. If the reference preferences match the objective the behaviour was actually generated by, the encoding fits and synergy is high. Shifting the reference away from that objective should disrupt the joint structure synergy measures. Three observations support this reading over alternatives. The response is selective: in the control and paired-mismatch conditions, 81 to 96 percent of the perturbation effect concentrates in the synergistic atom, while redundancy and unique information remain approximately stable (Section 4.2), which rules out generic shifts in joint information content. Identical agents produce different perturbation responses depending on which preferences they actually optimised for in the mismatch experiment (Section 4.1), isolating the encoding-behaviour relationship from any property of the underlying model. The response also inverts within the same agent between control and mismatch in the GovSim probe, with the outsider producing ratio 0.53 under sustainable preferences and 3.68 under extractive ones (Section 4.3). The signature is not that synergy is high or low, but that synergy responds to perturbation in the direction the encoding-fit hypothesis predicts.

Limitations. The method, as presented in our main experiments, requires that each agent's declared objective be available as a numeric score function, which restricts application to settings where preferences are not only verbalised. The estimator requires population-level data, because per-pair triplet counts within a single interaction are too low for reliable single-episode auditing. Bootstrap intervals are computed at the game level, which preserves within-game trajectory dependence but assumes games are independent draws, an assumption that may not hold when games share prompts or initial conditions in tightly templated benchmarks. The source-feature encoding is substrate-specific. Section 4.3 shows that the negotiation encoding does not transfer directly to GovSim's one-dimensional harvest decision, and the redesigned encoding preserves the mismatch signature but introduces residual response in the control. Finally, all experiments use ground-truth mismatch by construction. The signature has not been tested under mismatch arising from training pressure, in-context manipulation, or population dynamics.

Broader implications. Two implications follow. For practitioners, the framework offers a post-hoc audit that does not require model access or message content and adds no inference cost beyond the original interaction, because the perturbation is applied to the encoding rather than re-run on the agents. An organisation that retains behavioural logs together with the score sheets the agents were given can apply the sweep retroactively. For methodology, the result that the level of synergy carries no signal on its own, only its response to perturbation, argues for treating information-theoretic measurements of multi-agent behaviour as relational rather than absolute. What matters is whether the joint structure synergy measures is anchored to the objective the agent was supposed to be optimising.

Future directions. Three extensions follow from the limitations above. The substrate-specificity of the source-feature encoding suggests a learned or shared encoding as a research direction. A featurisation trained to preserve the mismatch signature across a multi-substrate calibration set would distinguish whether the encoding-behaviour relationship is a property of the method or a property of the particular features. Secondly, because the perturbation requires no additional model calls, a real-time variant running rolling sweeps over recent behaviour windows is feasible, and whether the signature is stable at that scale is an open empirical question. Finally, the framework currently operates on the encoding side of the encoding-behaviour relationship, while other behaviour-level audits intervene on the action side through paraphrase, role permutation, or message-level interventions.

6 Conclusion

Whether each agent in a multi-agent LLM system behaves consistently with its declared objective is a question that outcome-level metrics often cannot answer. We have shown that counterfactual perturbation of declared preferences, read through pairwise partial information decomposition, produces a behaviour-level signature that localises objective mismatch by agent, by preference dimension, and by time window in a multi-issue negotiation benchmark, and that the mismatch signal partially survives transfer to a categorically different substrate, albeit with a residual control-side response that reflects substrate-dependent encoding. We present the framework as proof of principle for a broader research direction on information-theoretic analysis of multi-agent behaviour.

LLM Usage Disclosure

The authors used Claude Opus 4.6 (Anthropic) as a writing assistant during manuscript preparation. The underlying research, including experimental design and interpretation of all results, was performed by the authors.

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A Agent-Specific Perturbation Profiles

This appendix reports the establishing experiment referenced in Section 4.1. We apply the counterfactual perturbation to three to four agents per game across all six released game configurations, sweeping α from 0 to 1 in 11 steps, yielding 19 agent-specific profiles in total. For each drift agent we normalise drift-pair synergy by its $\alpha = 0$ value and plot it against α; the ten same-game control pairs that do not contain the drift agent are unaffected by construction and remain flat. In every game the drift-pair curves fall as the perturbation grows (18–72% drop at the endpoint), and the response is predominantly synergistic (synergy accounts for 90–99% of the total absolute atom change). The steepness of the drop tracks each agent's stake in the negotiation, the total spread of its score sheet: high-stakes agents drop steeply, low-stakes agents drop shallowly. All six games use honest play under true preferences, so these profiles establish that the perturbation produces a selective, agent-specific, stakes-graded response, not that it tracks ground-truth misalignment; the latter is tested in Section 4.1.

B Cross-Model Replication

To check that the perturbation signature is not specific to one model family, we replicate the agent-level drift sweep on the cooperative game with both GPT-4 (32 games) and Llama-3-70B (33 games). Both models show the same selective response: the probed agent's drift pairs drop under perturbation while same-game control pairs stay flat (Figure 6).

C Atom Concentration

We summarise drift-pair atom means at the perturbation endpoints by condition in Table 1. The synergy fraction is computed as

$$\frac{|\Delta\mathrm{Syn}|}{\sum_X |\Delta X|}, \qquad \Delta X = X(\alpha = 1) - X(\alpha = 0),$$

where the sum runs over the four atoms. This is the fraction of total absolute atom motion attributable to synergy.

Table 1: Drift-pair atom means at perturbation endpoints by condition, and the share of total absolute atom motion concentrated in synergy. Control and paired-mismatch show strong concentration of the response in synergy. Single mismatch is the framework's expected null: the absolute change in synergy is small (−0.012 bits), and what motion remains distributes more evenly across atoms.

Condition	α	Red	Un_1	Un_2	Syn	syn-frac
Control	0.0	0.030	0.007	0.012	0.361	
	1.0	0.029	0.006	0.014	0.240	0.96
Paired mismatch	0.0	0.038	0.013	0.015	0.424	
	1.0	0.041	0.010	0.011	0.380	0.81
Single mismatch	0.0	0.057	0.026	0.020	0.549	
	1.0	0.053	0.017	0.023	0.538	0.43

The two conditions where the perturbation produces a meaningful synergy drop (control with $|\Delta\mathrm{Syn}| = 0.121$ and paired mismatch with $|\Delta\mathrm{Syn}| = 0.044$) show concentration ratios of 0.96 and 0.81 respectively. Single mismatch, where the probed agent's behaviour was generated under reversed preferences and is therefore already decoupled from the reference at $\alpha = 0$, shows a small absolute synergy change (0.012 bits) and a synergy fraction of 0.43. We read this as the expected pattern: the framework's synergy-concentration property holds when synergy is meaningfully responding to the perturbation, and does not hold when synergy is approximately flat because there is no response to concentrate.

D Robustness Checks

We verify that the headline localisation signal is preserved under variation in the four main design choices of the framework: source-feature encoding, target variable, binning resolution, and PID estimator.

Source feature encoding. The baseline encoding pairs `self_score` with `payoff_sensitivity` as the two-dimensional source. We test three alternatives: 1D self-score only (collapses to mutual information rather than PID), self-score rank in place of raw value, and self-score distance from the agent's threshold. The control-vs-paired-mismatch gap on the target issue is the relevant diagnostic. The 2D encoding recovers a gap of 0.231; 1D self-score-only collapses to 0.135 and loses the signal direction in two of the secondary issues; rank and distance encodings preserve direction with reduced magnitude (0.189 and 0.138). The 2D encoding is the only one that recovers the full agent-by-issue scan with the two ground-truth cells from Section 4.1 among the largest positive gaps.

Table 2: Source-encoding sensitivity. Control and paired-mismatch endpoint synergy ratios with cluster-bootstrap 95% CIs (1000 resamples). Gap = paired-mismatch ratio − control ratio. The 1D self-only encoding reverses the control direction (ratio > 1) and widens CIs markedly.

Encoding	Control ratio [95% CI]	Paired-mismatch ratio [95% CI]	Gap
`self_sens` (baseline)	0.665 [0.545, 0.768]	0.896 [0.810, 0.983]	+0.231
`self_only` (1D)	1.047 [0.262, 1.597]	1.182 [0.659, 2.084]	+0.135
`self_rank`	0.730 [0.618, 0.830]	0.919 [0.832, 1.008]	+0.189
`self_dist`	0.721 [0.623, 0.810]	0.859 [0.769, 0.930]	+0.138

Target variable. The baseline target is the non-pair welfare delta. We test three alternatives: pair welfare delta (own payoff change), total welfare delta (all six agents), and drift-only delta (perturbed agent's payoff change). Gaps remain in the same direction across all four choices (0.231, 0.304, 0.229, 0.286 respectively), indicating that the framework's localisation is not an artefact of the non-pair target specifically. We retain non-pair as the headline choice because it most clearly separates the framework's question (does the pair predict externalities on others) from a mechanical accounting of the pair's own payoffs.

Binning resolution. We sweep $B \in \{2, 3, 4, 5\}$ for the source dimensions and verify that the perturbation drop persists in the same direction across all four (Table 4). This sweep was run on the cooperative baseline rather than the 65-game split, so it confirms the drop is not a binning artefact but does not re-establish the control-vs-mismatch gap at each B. $B = 4$ is the headline choice.

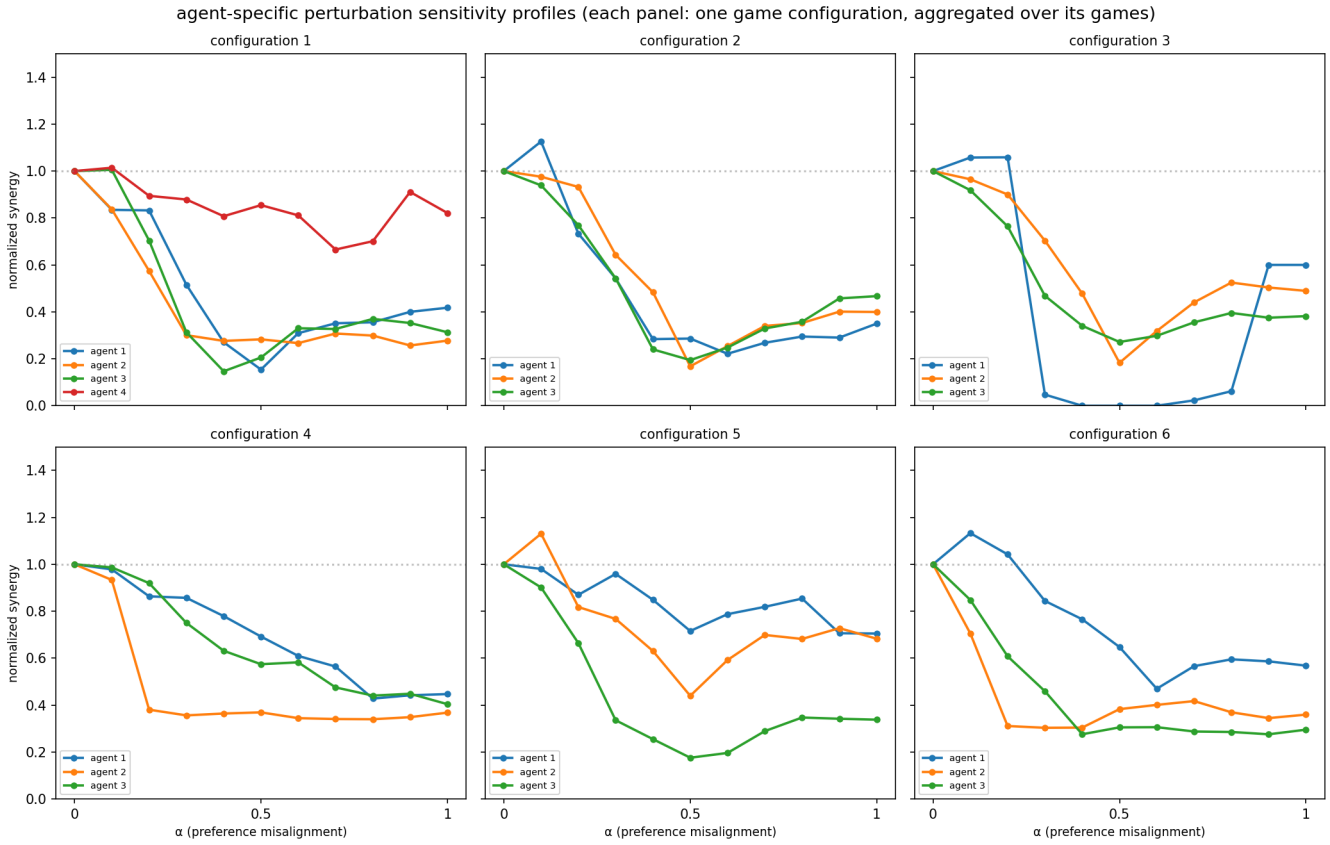


Figure 5: Agent-specific perturbation sensitivity profiles across all six game configurations. Normalised drift-pair synergy (÷ own $\alpha = 0$ value) versus perturbation level α, one line per drift agent. Curves fall 18–72% from reference to inverted preferences; same-game control pairs (not shown) remain flat. Drop steepness tracks each agent's score-sheet spread.

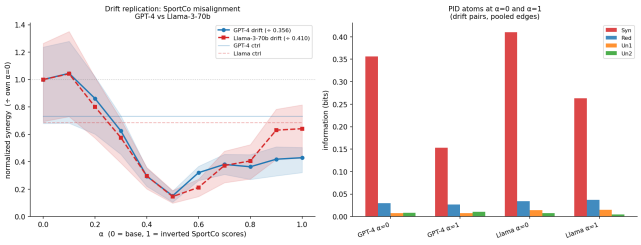


Figure 6: Cross-model replication. Normalised synergy versus α for the probed agent's drift pairs and control pairs, overlaid for GPT-4 (32 games) and Llama-3-70B (33 games). Both models show the same selective drop in drift pairs with flat control trajectories.

PID estimator: $I_{\min}$ vs BROJA.. The Williams-Beer $I_{\min}$ estimator used throughout the paper is known to over-attribute redundancy in some regimes. BROJA [2] addresses this through a constrained convex optimisation at substantially higher computational cost. We rerun the agent-level experiment under both estimators on pooled drift-pair triplets at the perturbation endpoints.

Table 3: Target-variable sensitivity. Same format as Table 2. All four target definitions preserve the direction and sign of the gap.

Target	Control ratio [95% CI]	Paired-mismatch ratio [95% CI]	Gap
non_pair (baseline)	0.665 [0.545, 0.768]	0.896 [0.810, 0.983]	+0.231
pair	0.642 [0.552, 0.761]	0.946 [0.847, 0.999]	+0.304
total	0.655 [0.538, 0.767]	0.884 [0.787, 0.975]	+0.229
drift_only	0.673 [0.567, 0.767]	0.959 [0.852, 1.020]	+0.286

The within-row $I_{\min}$-to-BROJA agreement within 4% across all conditions establishes that the perturbation response, on the pooled aggregation tested here, is not an artefact of the estimator's redundancy attribution.

E Temporal Localisation: Implementation Details

Intervention. The probed agent's preference sheet is silently swapped at round 12 of 24. Rounds 1–12 use target-issue scores [35, 29, 20, 0]; rounds 13–24 use [0, 20, 29, 35], the reversed ranking. No other agent is modified and no system message announces the

Table 4: Binning-resolution sensitivity (probed-agent drift, cooperative condition, $N_{\text{target}} = 3$ fixed). The sweep was run on the cooperative baseline (F1) rather than on the 65-game collusion split; stored data contains only the single-condition drop ratio, not the control-vs-mismatch gap. All four values of B produce a consistent drop (ratio < 1) in the same direction.

B	Syn α=0	Syn α=1	Drop ratio	Ctrl Syn α=0
2	0.032	0.017	0.539	0.022
3	0.147	0.086	0.586	0.128
4	0.377	0.157	0.418	0.269
5	0.463	0.222	0.480	0.408

Table 5: Endpoint synergy ratio under $I_{\min}$ and BROJA, by condition. Agreement within 4% across all three conditions. Ratios are computed on pooled drift-pair triplets (all triplets combined, PID run once), whereas headline ratios in the main body use mean-of-per-pair aggregation (PID per pair, ratios averaged); the two weightings differ, so absolute ratios are not identical. The relevant quantity here is the within-row $I_{\min}$-to-BROJA agreement, which holds within 4% in every condition.

Condition	$I_{\min}$	BROJA	Syn_0 ($I_{\min}$)	Syn_1 ($I_{\min}$)	Syn_0 (BROJA)
Control	0.663	0.640	0.149	0.099	0.145
Paired mismatch	0.963	0.970	0.167	0.161	0.164
Single mismatch	0.947	0.957	0.268	0.254	0.262

change; the probed agent discovers the discontinuity through its own bookkeeping.

Per-window bin edges. The two windows produce different probed-agent behaviour distributions by design. Pooling them when fitting bin edges suppresses the perturbation response (a pooled-edges pilot returned Window A ratio $\approx$ 0.9 where per-window edges recover 0.59). Edges are therefore fitted independently within each window at $\alpha = 0$.

Pre-registered predictions. Window A: $[0.40, 0.70]$, observed 0.588. Window B: $[0.85, 1.15]$, observed 1.001. Gap ≥ 0.20, observed 0.413. All three in range.

Table 6: Per-window synergy ratios and bootstrap CIs.

Window	Ratio	CI low	CI high	Syn α=0	Syn α=1
A (pre-update)	0.588	0.570	0.800	0.708	0.416
B (post-update)	1.001	0.825	1.197	0.291	0.291

F Diagnosed Negative Environment: Auction

To test whether the framework extends beyond multi-issue negotiation, we apply counterfactual preference perturbation to a competitive English auction. Four LLM bidders (GPT-4.1-mini, temperature 0.9) with private valuations bid sequentially over eight items per game. Bidder 1 receives a single modification across three conditions: *control* (50 games, unmodified), *swap* (50 games, top-two and bottom-two valuations swapped), and *shill* (1 pilot game, instructed to inflate prices on two pre-sampled high-value items without intending to win them). The perturbation interpolates Bidder 1's reference valuations toward a rank-inverted copy, parallel to the negotiation design. Sources use exit-ratio (exit price over reference valuation, clipped to $[0, 3]$) and cumulative surplus over budget; the pair target is the non-pair surplus delta.

Table 7: Normalised mean drift-pair synergy at each α step, per window (÷ own α=0 baseline; raw bits in parentheses). Window A falls monotonically after $\alpha \approx 0.6$; Window B oscillates without trend.

α	Window A		Window B	
	ratio	(bits)	ratio	(bits)
0.0	1.000	(0.708)	1.000	(0.291)
0.1	0.974	(0.689)	1.093	(0.318)
0.2	0.869	(0.615)	1.043	(0.303)
0.3	0.853	(0.604)	0.908	(0.264)
0.4	0.885	(0.626)	0.825	(0.240)
0.5	0.835	(0.591)	0.888	(0.258)
0.6	0.825	(0.584)	0.948	(0.276)
0.7	0.592	(0.419)	0.938	(0.273)
0.8	0.625	(0.442)	0.997	(0.290)
0.9	0.597	(0.423)	0.874	(0.254)
1.0	0.588	(0.416)	1.001	(0.291)

Table 8: Drift-pair mean atom values at perturbation endpoints, per window. Synergy fraction is $|\Delta\text{Syn}|/\sum_X |\Delta X|$. Window A concentrates 64% of total atom motion in synergy; Window B has near-zero synergy response (0.6%) with no directional trend in the remaining atoms.

Window	α	Red	Un_1	Un_2	Syn	syn-frac
A	0.0	0.141	0.055	0.019	0.708	
	1.0	0.084	0.010	0.077	0.416	0.645
B	0.0	0.040	0.008	0.032	0.291	
	1.0	0.057	0.027	0.015	0.291	0.006

The behavioural manipulations land at the auction-outcome level. Bidder 1 wins the highest-valued item in 62% of control games and 0% of swap games (Table 9, full data in `sanity/raw_behaviour_check.md`). Exit-ratios on top-ranked items shift from 0.60 (control) to 0.26 (swap), and on bottom-ranked items from 0.87 to 2.46, confirming that the swap manipulation reverses Bidder 1's revealed preferences on observable behaviour.

The behavioural shift is therefore present and large. The PID perturbation response, however, does not recover the expected localisation signature. We pre-registered drop-ratio predictions for each (bidder, condition) cell (Appendix F, auction predictions); the per-bidder ratios under the exit-ratio encoding diverged from these predictions in a pattern that does not isolate Bidder 1 or the swap and shill conditions from control.

We interpret this as a scope condition of the framework rather than a method failure. Two properties of the auction substrate

Table 9: Bidder 1 win fraction by item value rank under reference valuations (rank 0 = highest-value item). The swap condition produces the expected inverted pattern at the outcome level. Shill data is from one pilot game only.

Rank	Control	Swap	Shill
0	0.62	0.00	1.00
1	0.44	0.00	0.00
6	0.06	0.74	0.00
7	0.02	0.94	0.00

Table 10: Per-bidder endpoint synergy ratio (drop ratio = $\mathrm{Syn}(\alpha{=}1)/\mathrm{Syn}(\alpha{=}0)$) under exit-ratio source encoding, real-mode valuation. Bidder 1* is the modified agent in swap and shill conditions. Values < 1 indicate a synergy drop; > 1 indicates synergy increase under perturbation. Shill column is from one pilot game and is not reliable for inference.

Bidder	Control	Swap	Shill†
Bidder 1*	2.163	0.733	1.274
Bidder 2	0.542	0.667	1.089
Bidder 3	0.626	0.699	2.559
Bidder 4	0.664	0.676	1.000

*Modified agent in swap and shill conditions. †One pilot game; not reliable for inference.

likely contribute. The action space is one-dimensional per item (a single bid or withdrawal), so the framework's two-feature source encoding compresses behaviour that the negotiation encoding distributes across distinct dimensions. The decision horizon is also short relative to the negotiation benchmark (eight items versus twenty-four rounds with five issues), producing fewer triplets per pair per game (seven transitions for items vs. ≈ 21 for negotiation). Sparse joint tables under the two-dimensional source $\times$ ternary target ($N_{\mathrm{BINS}} = 2$, joint dimension $4 \times 4 \times 3 = 48$ cells) are estimated from approximately $50 \times 7 = 350$ triplets per pair per condition, roughly half the negotiation data volume on a denser joint table. We report this here as a documented scope condition. A successful auction application would require either a higher-dimensional source encoding that captures bid trajectory rather than item-level summary, or substantially more games to populate the joint table.

G Cluster Bootstrap Implementation

Algorithm. All confidence intervals in the paper are computed via a cluster bootstrap with the game as the resampling unit. For each of B=1000 iterations, n games are drawn with replacement from the n available games. PID is re-computed on the pooled triplets from the resampled set at α=0 and α=1, and the synergy ratio (or atom delta) is recorded. The 2.5th and 97.5th percentiles across iterations yield the 95% CI.

Why games, not triplets. Resampling individual triplets would underestimate variance because triplets within a game share a common negotiation context. Two reasons justify the game-level choice:

(1) Utterances in later rounds are conditioned on earlier ones in the same game; triplets are not exchangeable within a game.
(2) Bin edges are fitted once across all triplets in a condition; resampling triplets while holding edges fixed would conflate sampling uncertainty with discretisation variance.

Game-level resampling preserves within-game dependence and supports inference over the population of games.

Trim threshold. Iterations where $S(\alpha{=}0) < 10^{-9}$ bits are discarded to avoid degenerate normalised ratios. This threshold was not triggered in any headline experiment: the minimum observed $S(\alpha{=}0)$ is approximately 0.148 bits (GovSim mismatch, the outsider).

Table 11: Bootstrap configuration per experiment and condition. Max dropped iterations is 0 everywhere because $S(\alpha{=}0) \gg 10^{-9}$ throughout; this value is not stored in results JSONs and was verified from script output.

Experiment	Condition	n_{games}	n_{boot}	Trim	Max dropped
Agent-level localisation	Control	25	1000	10^{-9}	0
	Paired mismatch	25	1000	10^{-9}	0
	Single mismatch	15	1000	10^{-9}	0
Temporal localisation	Window A	25	1000	10^{-9}	0
	Window B	25	1000	10^{-9}	0
GovSim	Control	30 (×5)	1000	10^{-9}	0
	Mismatch	22–30 (×5)	1000	10^{-9}	0
Issue/agent scan	Control	25	1000	10^{-9}	0
	Collusion	25	1000	10^{-9}	0

H GovSim Cross-Environment: Implementation and Full Results

This appendix documents implementation choices for Section 4.3 and reports the full perturbation-ratio table for all five agents in both conditions. The source encoding used in GovSim is substrate-specific and differs from the negotiation framework in ways that we disclose here because the choice is methodologically load-bearing for the substrate-dependence finding in Section 5.

H.1 Outcome-Level Results

We ran 30 control and 30 mismatch games on Llama-3-70B served via vLLM, unquantised, with `max_tokens` = 300 and CUDA graph compilation disabled. The perturbation analysis requires a valid synergy baseline at $\alpha = 0$; this holds for all 30 games in every persona except the outsider in mismatch, where 22 games qualify, which sets the 22–30 range in Table 11. In mismatch, the outsider enters at round 3 with declared preferences that favour maximal extraction. At the outcome level the manipulation lands. Mean commons survival drops from 11.7 rounds in control to 5.8 rounds in mismatch, and the outsider's mean harvest rises from 5.6 to 29.5 fish per round. Outcome metrics therefore distinguish the conditions in this substrate, which establishes that the misalignment is behaviourally manifest before the perturbation analysis is applied.

H.2 Source Encoding

GovSim's per-round harvest decision is one-dimensional, which precludes the direct two-feature encoding used for negotiation. Reaching a feature set that produced a meaningful localisation signal required non-trivial iteration on the encoding design. We report the encoding actually used and the choices it embeds.

For agent i at round t with raw harvest h_i^t and per-capita sustainable share $s_i^t = \max\big(\text{pool_before}^t/(2n_t),\ 1.0\big)$, the source features are

$$f_{1,i}^t(\alpha) = \frac{h_i^t}{(1-\alpha)\, s_i^t + \alpha \cdot 30}, \qquad f_{2,i}^t(\alpha) = \left|f_{1,i}^t(\alpha) - f_{1,i}^{t-1}(\alpha)\right|. \quad (6)$$

The interpolated target sweeps between the sustainable share ($\alpha = 0$, the aligned reference) and a fixed extractive target of 30 fish per round per agent ($\alpha = 1$, three times the sustainable share at $n = 5$, pool = 100). The pair target is the non-pair harvest delta, parallel to the negotiation framework and α-independent.

Two choices in this encoding deviate from negotiation and merit explicit disclosure. First, the second source feature is the absolute round-over-round change in f_1 rather than an independent behavioural dimension. GovSim provides no analogue to negotiation's payoff sensitivity because the per-round action space is scalar; we differenced f_1 to recover a second dimension. Second, the extractive target was fixed at 3× the sustainable share rather than at the literal maximum (the full pool). The 3× choice preserves dynamic range at $\alpha = 1$ so that the encoding does not saturate; the literal maximum produced degenerate distributions in pilot runs. Bin edges are fitted per (condition, drift agent) pooled across all α values to avoid discretisation drift along the sweep.

H.3 Full Perturbation Ratios

Table 12 reports the synergy ratio $r = \text{Syn}(\alpha = 1)/\text{Syn}(\alpha = 0)$ with cluster-bootstrap 95% CIs for each of the five agents under each condition, together with the triplet pool size at $\alpha = 0$.

Table 12: GovSim perturbation ratios, cluster-bootstrap 95% CIs, $n = 1000$ resamples. The outsider is the extractive agent in the mismatch condition and an ordinary aligned resident in control. The smaller triplet pool for the outsider in mismatch reflects the earlier commons collapse in those games.

Condition	Agent	Ratio	95% CI	n triplets ($\alpha = 0$)
Control	resident 1	0.537	[0.391, 0.805]	1284
	resident 2	1.082	[0.835, 1.501]	1284
	resident 3	1.756	[1.235, 2.434]	1284
	resident 4	1.427	[1.110, 1.891]	1284
	outsider	0.527	[0.401, 0.799]	1284
Mismatch	resident 1	0.394	[0.273, 0.602]	482
	resident 2	0.949	[0.739, 1.221]	482
	resident 3	0.900	[0.742, 1.261]	482
	resident 4	0.486	[0.364, 0.819]	482
	outsider	3.682	[1.361, 5.513]	212

The headline observations are consistent with the main-body discussion: under mismatch, the outsider's interval is the only one excluding 1.0 in the upward direction, which is the expected localisation signature for the outsider. In control, residents 3 and 4 produce intervals that exclude 1.0 in the wrong direction, which we read as substrate-dependence of the feature encoding rather than a method failure. The same intervals do not rise under mismatch (resident 3 from 1.76 to 0.90; resident 4 from 1.43 to 0.49), which is the pattern one would expect if substrate noise is dominated by a genuine misalignment signal when one is present.

I Scalar-Measure Comparison

This appendix accompanies the comparison in Section 4.2. Across all six game configurations we run the same perturbation sweep and plot the normalised response of PID synergy, the best-response gap, and the Nash bargaining solution distance. PID synergy is uniform in shape across configurations, whereas the best-response gap and NBS distance take variable functional forms, confirming that the clean monotonic response is specific to synergy.

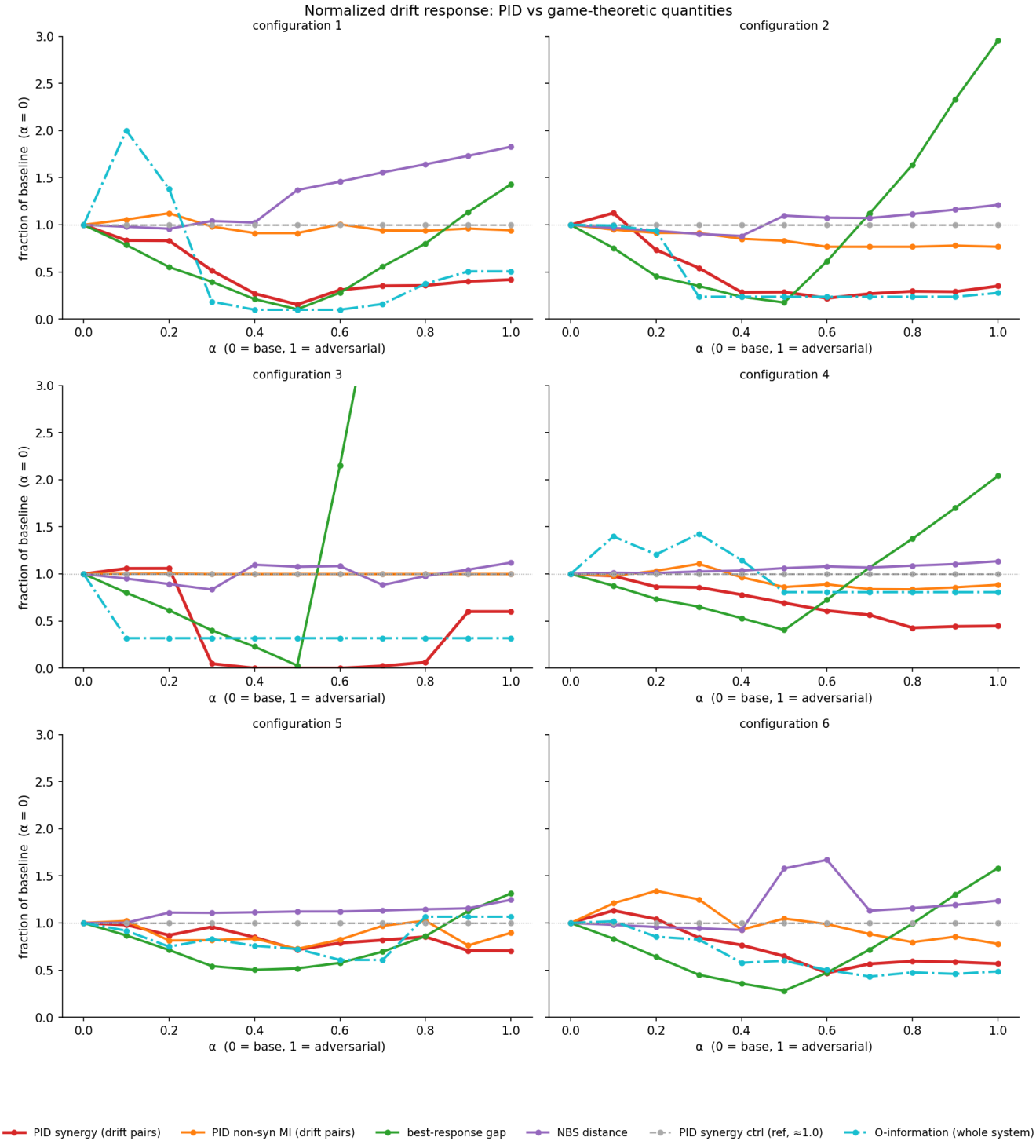


Figure 7: PID synergy compared to best-response gap and NBS distance under the same perturbation sweep. Each panel is one game configuration; within a panel, every line is one measure's normalised response across the α sweep. PID synergy (drift pairs) is uniform in shape across configurations, while the best-response gap and NBS distance vary.